

Computer Hacking Forensic Investigator (CHFI)

Version 11

Lab Report

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Course Name: Computer Hacking Forensic Investigator (CHFI V11)

Report Title: CHFI Lab Report (Module 11)

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Lab Scenario

When forensic investigators come across cases of dark web crimes, they generally do not locate any traces of criminal activity on normal browsers meant for daily use such as Google Chrome Mozilla Firefox, Microsoft Edge/Internet Explorer, etc. For perpetrating dark web crimes, criminals prefer the Tor Browser as it provides them with a high level of anonymity. In such instances, investigators need to examine the suspect machine for Tor Browser artifacts to help them solve such cases.

Tor Browser artifacts can be retrieved by examining the RAM dump of the suspect machine. In case Tor Browser has been removed/uninstalled from a system after committing the crime, investigators can discover its related artifacts by analyzing its prefetch file on a Windows machine.

Lab Objectives

The objective of this lab is to help you understand dark web forensics techniques. The tasks include the following:

- Discovering Tor Browser activity on a Windows machine
- Examining RAM dumps to discover Tor Browser artifacts

Overview of Dark Web Forensics

The use of dark web forensics is necessary because of an increase in illegal activities such as drugs and weapons trafficking, selling credit card details and other private information related to individuals, etc. that are facilitated through the internet. Dark web forensics helps investigators gather evidence against the perpetrators of such dark web crimes so that they can be prosecuted in a court of law.

Lab Tasks

Recommended labs to assist you in dark web forensics:

- Detect Tor Browser Activity on Windows Machine
- Analyze RAM Dumps to Retrieve TOR Browser Artifacts

Lab 1: Investigate a Suspicious Email

Lab Scenario

During a search and seize operation connected with a case of an Internet-based fraud in which credit card information and bank details of several individuals were stolen and sold, law enforcement authorities got hold of a few computers belonging to the suspects. During the forensic investigation of the seized computers, the investigators discovered that the suspects had tried to cover their tracks by removing all the tools/applications they had used on the systems. The investigators decided to examine the prefetch files on the system to determine the application/program that might have been used by the suspects to commit crimes. While searching through the prefetch files, the investigators located a prefetch file for Tor Browser, which indicates that this might be a case of dark web crime. How should the investigators proceed to be able to analyze the Tor Browser artifacts in this case?

As a forensic investigator, you must know how to analyze prefetch files pertaining to Tor Browser using the right tool(s).

Lab Objectives

Tor Browser is based on the Mozilla Firefox browsing application that works on the concept of Onion Routing. It allows users to access the dark web and carry out criminal/illegal activities anonymously.

In this lab, you will learn how to retrieve Tor Browser activity from a Windows machine by:

- Examining Windows prefetch files
- Monitoring network connections using netstat command

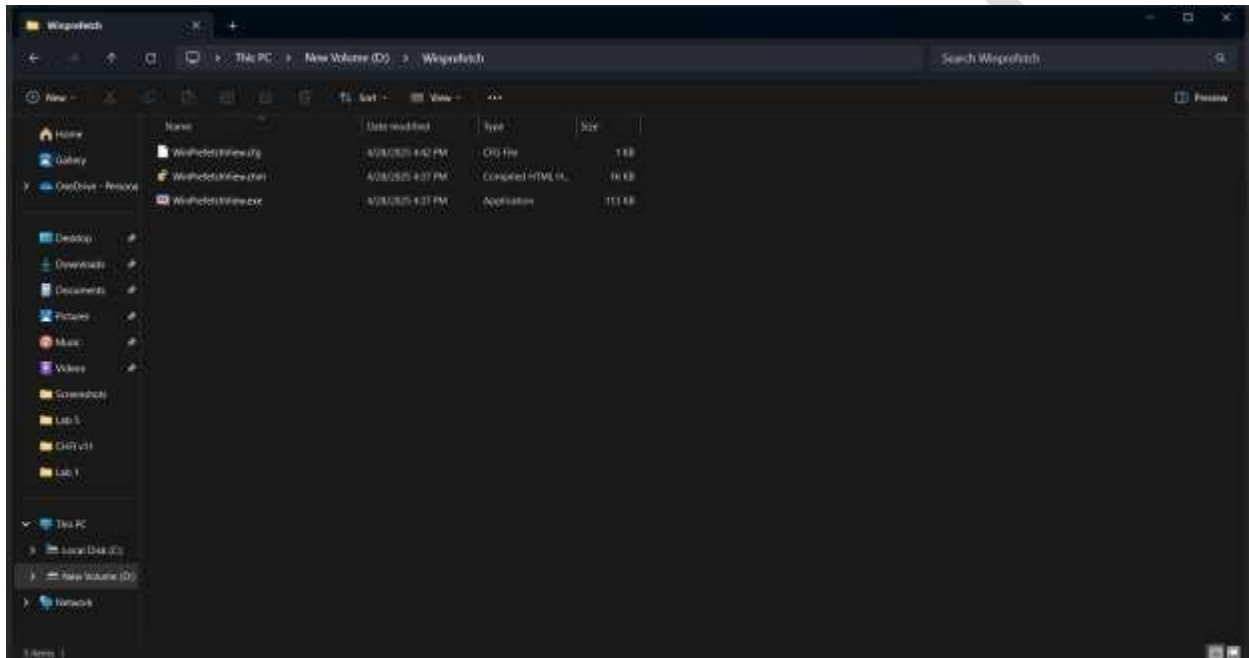
Overview of the Lab

This lab familiarizes you with the process of determining whether Tor Browser has been used on a suspect system with the help of WinPrefetchView, an application that helps you find the prefetch files pertaining to various programs that have been installed on a system. Even if a program is deleted/uninstalled from a system, the prefetch file related to it is still likely to remain on the system, which provides evidence for that program's execution on the system.

Make sure that Real Time Protection is disabled in the Windows 11 virtual machine (if it is running) before beginning this lab.

Lab Tasks

1. Navigate to D:\Winprefetch and launch the software WinPrefetchView.exe as shown in screenshot:



2. The GUI of WinPrefetchView.exe will look like shown in screenshot below:

Filename	Created Time	Modified Time	File Size	Process EXE	Process Path	Run Counter	Last Run Time	Missing Pre...
TOR-BROWSER-WINDOWS-X86-PORTABLE-1.4.1	5/3/2025 2:07:25 PM	5/3/2025 2:07:25 PM	47,524	TOR-BROWSER-WINDOWS-X86-PORTABLE-1.4.1	C:\Users\user\AppData\Local\Tor\TOR-BROWSER-WINDOWS-X86-PORTABLE-1.4.1	1	5/3/2025 2:07:25 PM	No

Filename	Full Path	Device Path	Network
18BFT	C:\Windows\System32\18BFT.dll	(\\VOLNAME)\18BFT.dll	24
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	8
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	10
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	7
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	43
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	1
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	2
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	27
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	22
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	25
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	34
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	18
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	26
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	19
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	25

5. The Properties window will now appear, displaying the metadata related to the browser such as Filename, Created Time, Modified Time, Last Run Time (the timestamp when the browser was last run), Run Counter (number of times the browser was executed), and Tor Browser execution directory (Process Path).

Filename	Created Time	Modified Time	File Size	Process EXE	Process Path	Run Counter	Last Run Time	Missing Pre...
TOR-BROWSER-WINDOWS-X86-PORTABLE-1.4.1	5/3/2025 2:07:25 PM	5/3/2025 2:07:25 PM	47,524	TOR-BROWSER-WINDOWS-X86-PORTABLE-1.4.1	C:\Users\user\AppData\Local\Tor\TOR-BROWSER-WINDOWS-X86-PORTABLE-1.4.1	1	5/3/2025 2:07:25 PM	No

Filename	Full Path	Device Path	Network
18BFT	C:\Windows\System32\18BFT.dll	(\\VOLNAME)\18BFT.dll	24
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	8
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	10
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	7
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	43
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	1
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	2
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	27
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	22
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	25
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	34
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	18
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	26
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	19
ADAPR32.DLL	C:\Windows\System32\ADAPR32.dll	(\\VOLNAME)\ADAPR32.dll	25

6. From the above screenshot, it is observed that Tor Browser has been executed 1 time on the machine (the Run Counter value varies depending on the number of times you

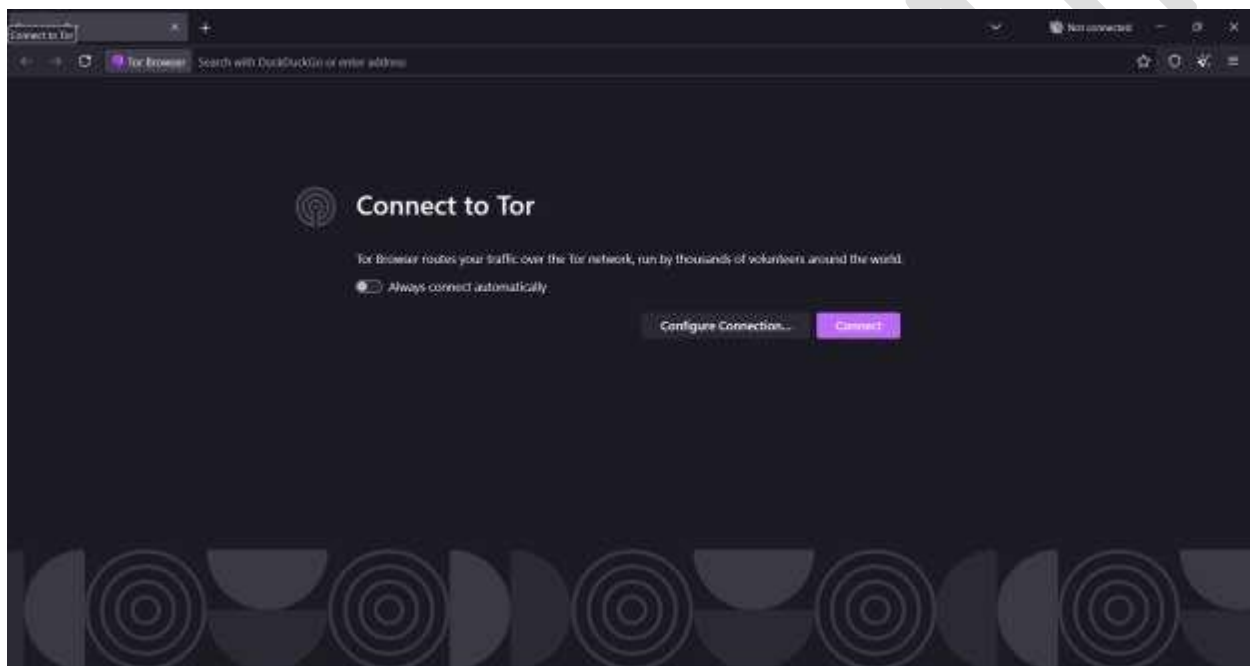
launch and access Tor Browser). The timestamps retrieved from the application help an investigator construct a timeline analysis of the incident.

WinPrefetchView application can fetch Tor Browser activity on the machine even when it is uninstalled/deleted from the machine.

7. Click OK in Properties window, and close the WinPrefetchView window.

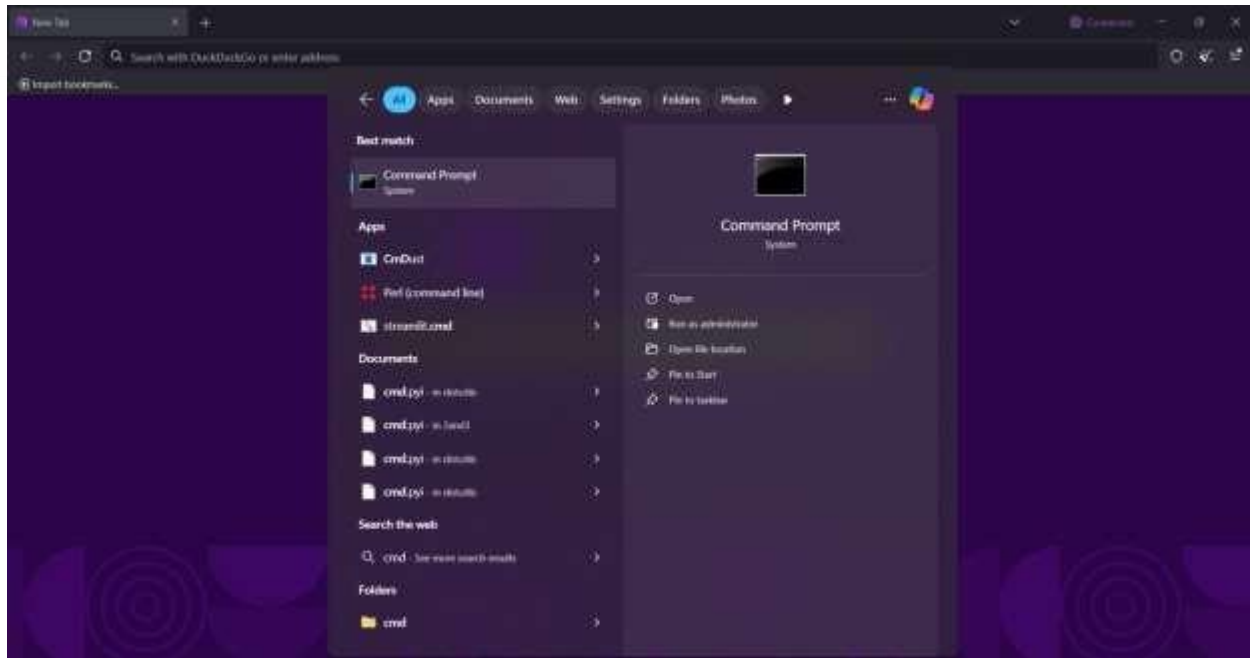
8. Double click on Tor Browser shortcut to run Tor browser and click on Connect and minimize the Tor browser window.

Ignore any errors if Tor Browser is unable to establish a Tor network connection.

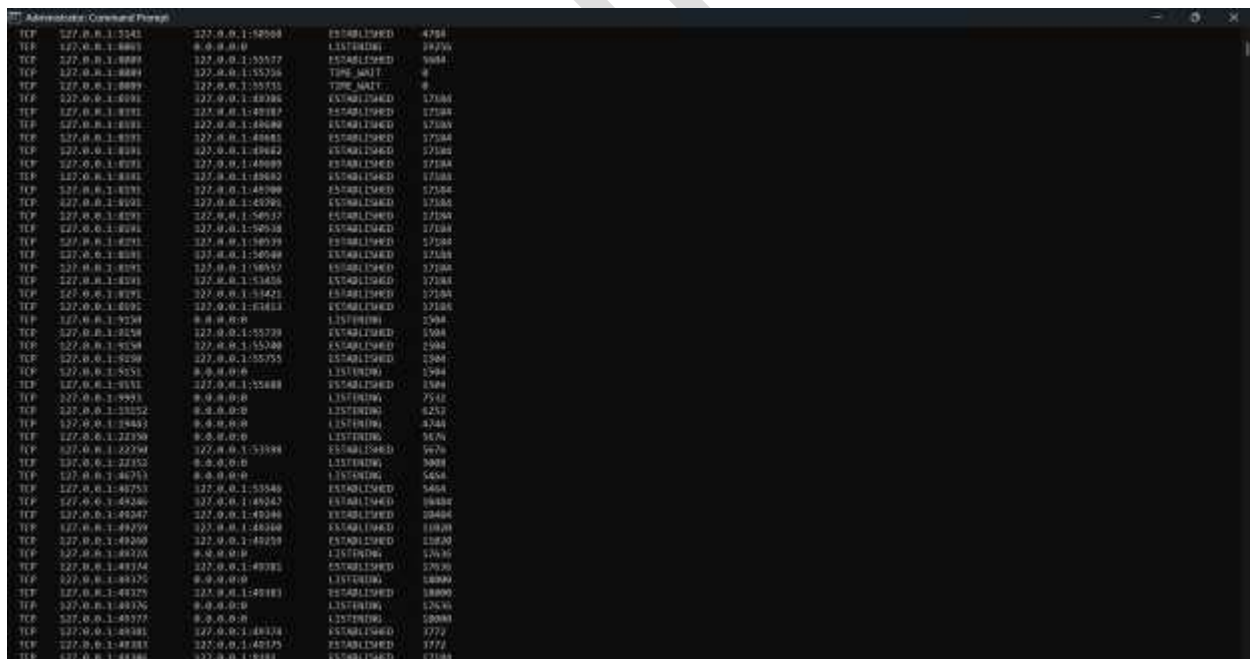


9. Click on Search icon and type cmd and select Run as administrator to run Command Prompt window.

If a User Account Control pop-up appears, click Yes to continue.



10. The netstat command in command prompt helps in determining Tor Browser activity on the suspect machine. If Tor successfully established a connection, then launching a command prompt with administrator privileges and running the netstat and command lists all active connection results as shown in the following screenshots:



When Tor Browser is installed on a Windows machine, it uses port 9150/9151 for establishing connection via Tor nodes.

The connection state ESTABLISHED means the socket has an established connection and LISTENING means the socket is waiting for a connection.

In case the Active Connections state is displayed as Time_Wait, it means that the Tor Browser application has been closed.

The result might differ if the browser has been closed a while ago. You might not see the ports 9150/9151 listed after a certain period of time lapse (for closed browser state).

11. In this manner, you can detect Tor Browser activity on a Windows machine.

12. Close all open windows.

Lab 2: Analyze RAM Dumps to Retrieve Tor Browser Artifacts

Lab Scenario

Forensic investigators have seized a computer belonging to a drug trafficker who is suspected of expanding his drug smuggling network through the dark web. During the investigation, it was found that the suspect had been using Tor Browser on his system to engage in drug trafficking and its expansion. To extract more information on the suspect's activities related to drug trafficking, investigators need to analyze the RAM dump of his system so that it reveals all his activities on Tor Browser. The artifacts obtained from the RAM dump can help the investigators extract evidence that can be used to prosecute the suspect.

As a forensic investigator, you must know how to analyze the RAM dump of a suspect machine and retrieve Tor Browser artifacts.

Lab Objectives

The memory dump collected from a suspect machine not only contains artifacts related to the browser, but also all the activities that occurred on it. Analyzing RAM dump can help investigators find all details pertaining to the activities that an attacker has performed on the system using the Tor Browser.

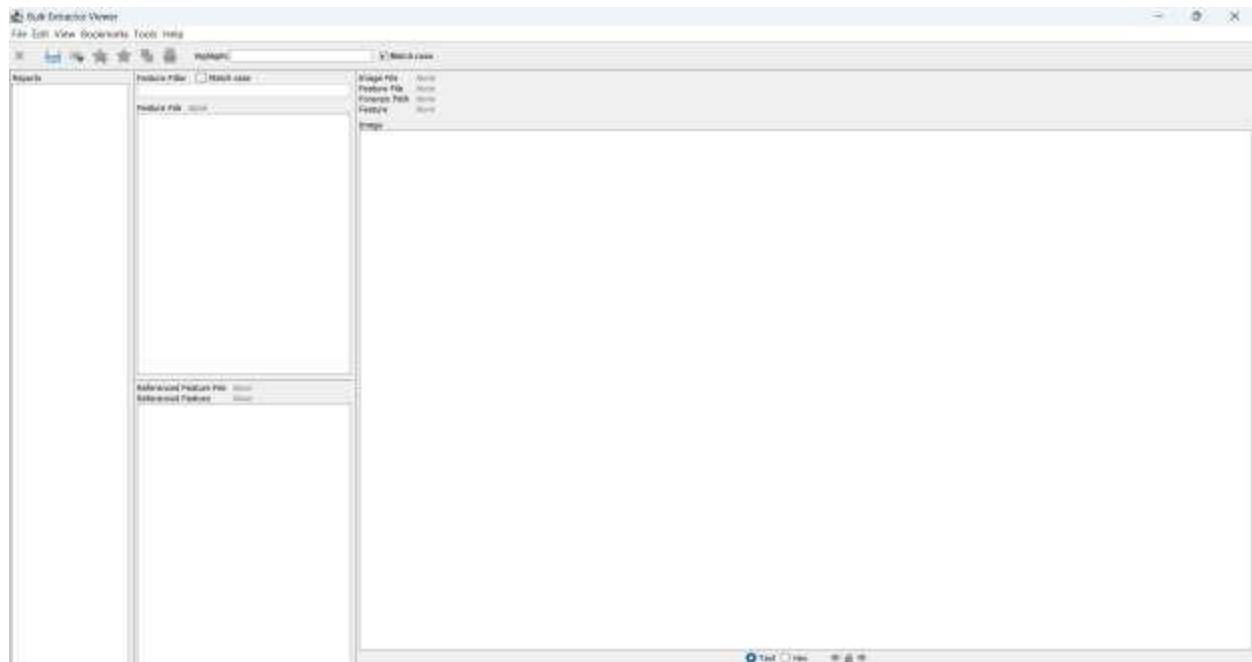
The objective of this lab is to help you learn how to examine a RAM dump and recover potential artifacts pertaining to Tor Browser using the Bulk Extractor tool.

Overview of the Lab

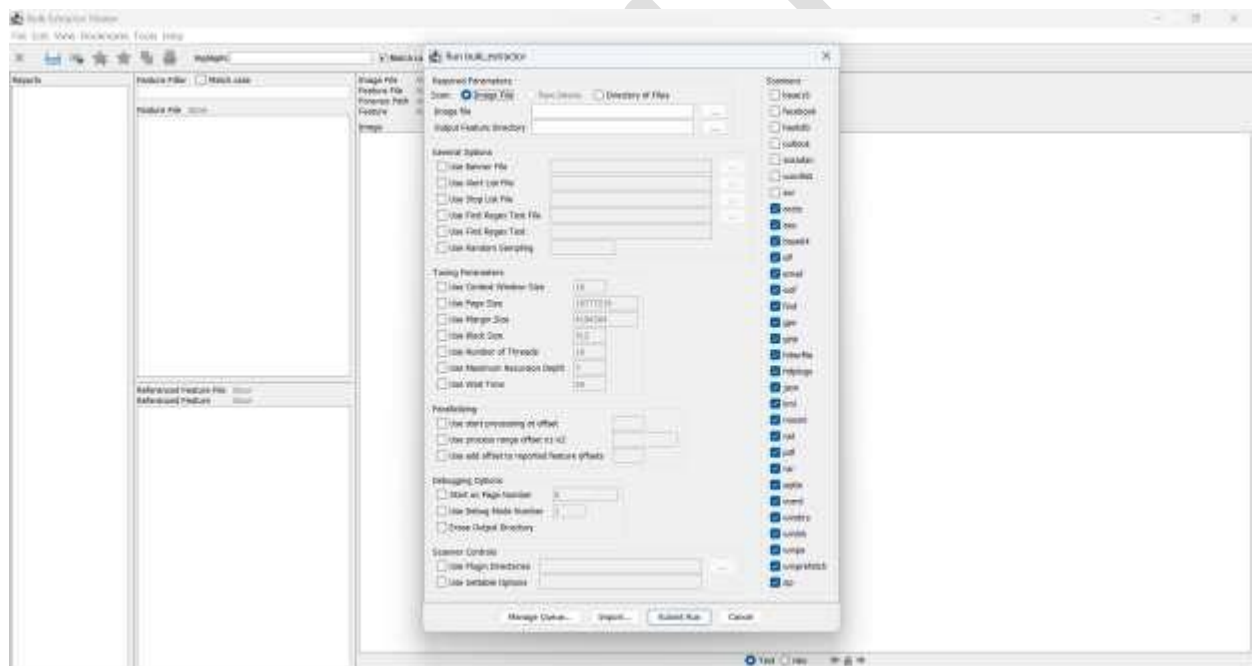
This lab familiarizes you with the process of analyzing a RAM dump containing Tor Browser artifacts with the help of Bulk Extractor.

Lab Tasks

1. Launch the Bulk Extractor Viewer and the GUI of the software look like screenshot below:

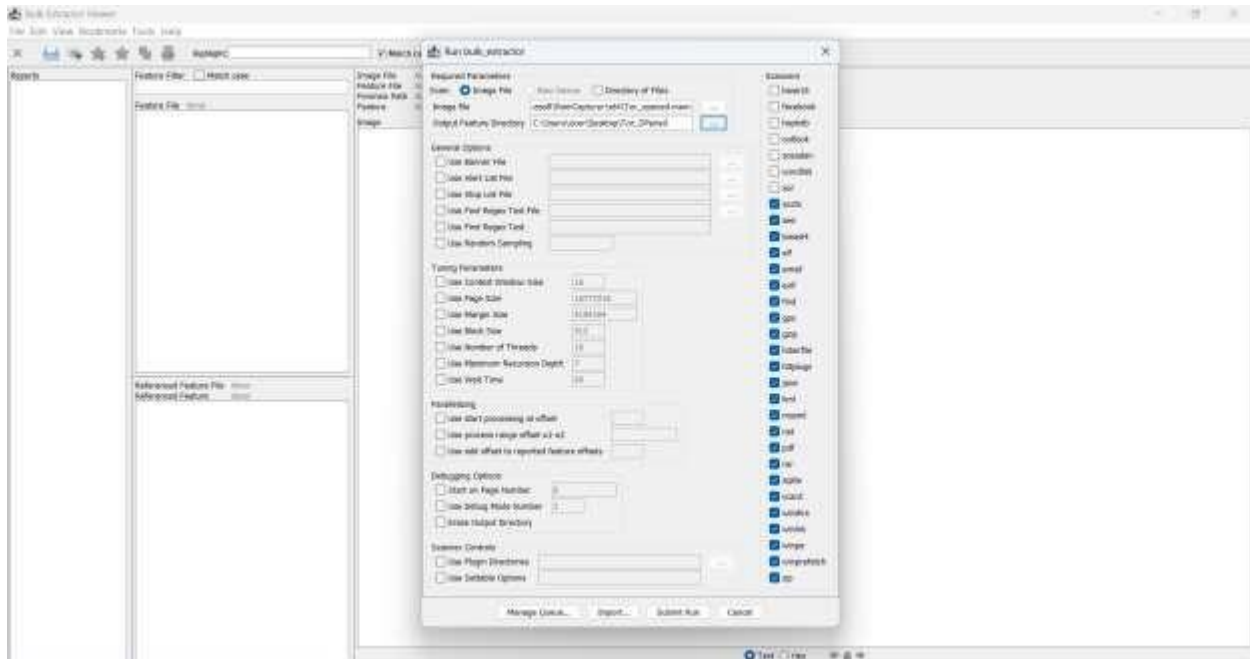


2. Now, the Run bulk_extractor window will open, as shown in the screenshot:

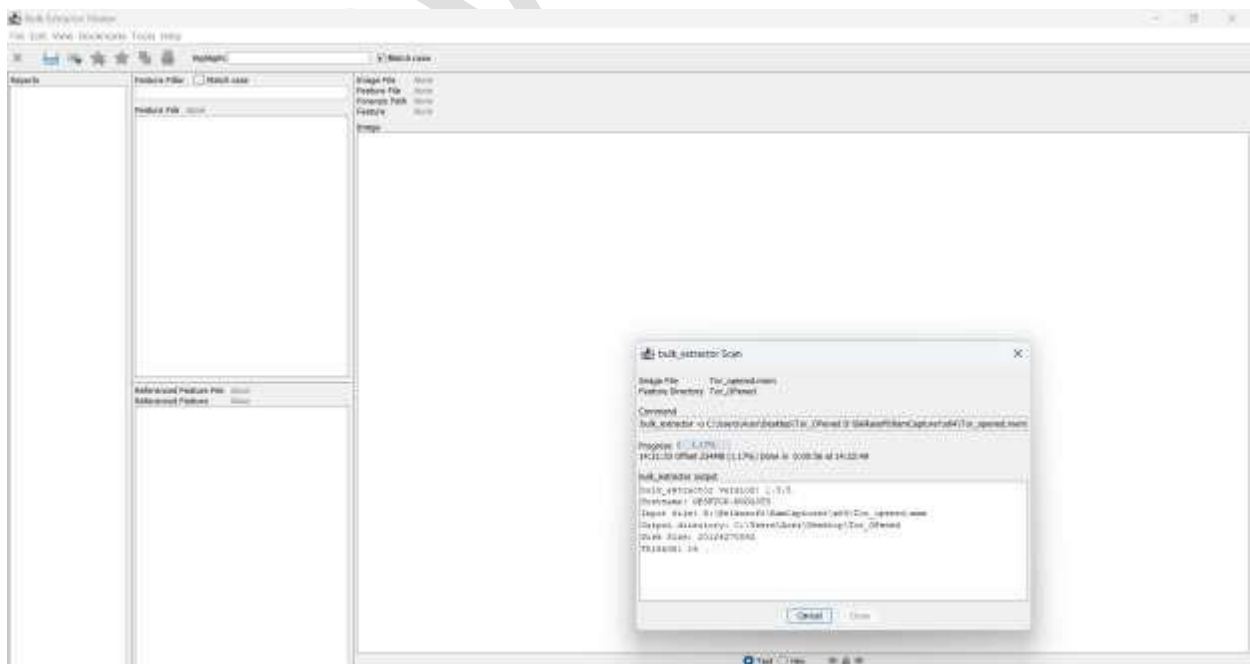


3. Now, we need to use the ellipsis buttons to browse the Image file and the Output Feature Directory, as indicated in the screenshot below:

4. We have provided the Image File and Output Feature Directory, and their paths will be displayed in their respective fields, as shown in the screenshot below. Now, ensure that all options under the Scanners section are checked and then click Submit Run, as highlighted in the screenshot:

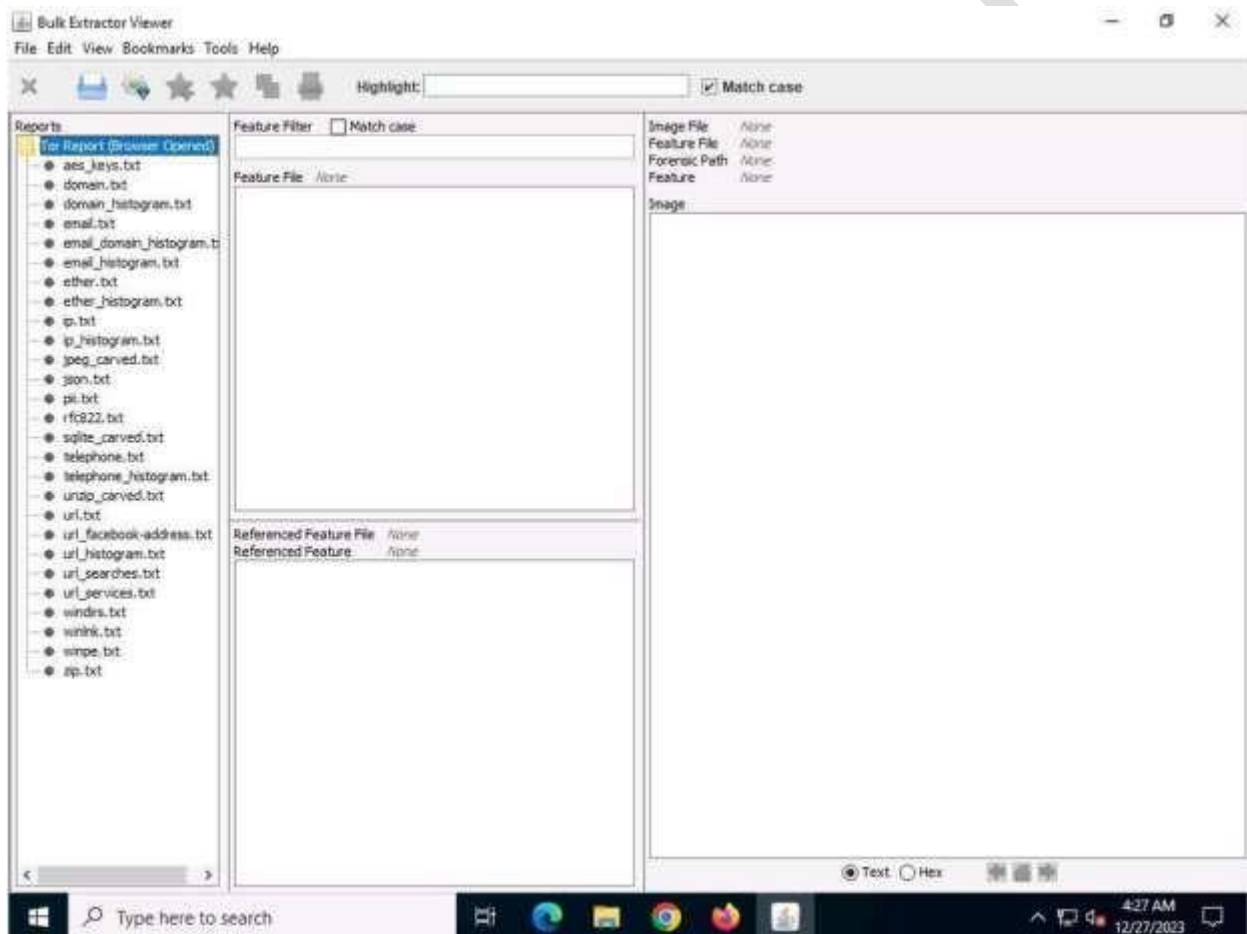


5. The bulk_extractor Scan window appears, where the input file is scanned. The progress of the scan and case creation can be seen in the window, as shown in the following screenshot:

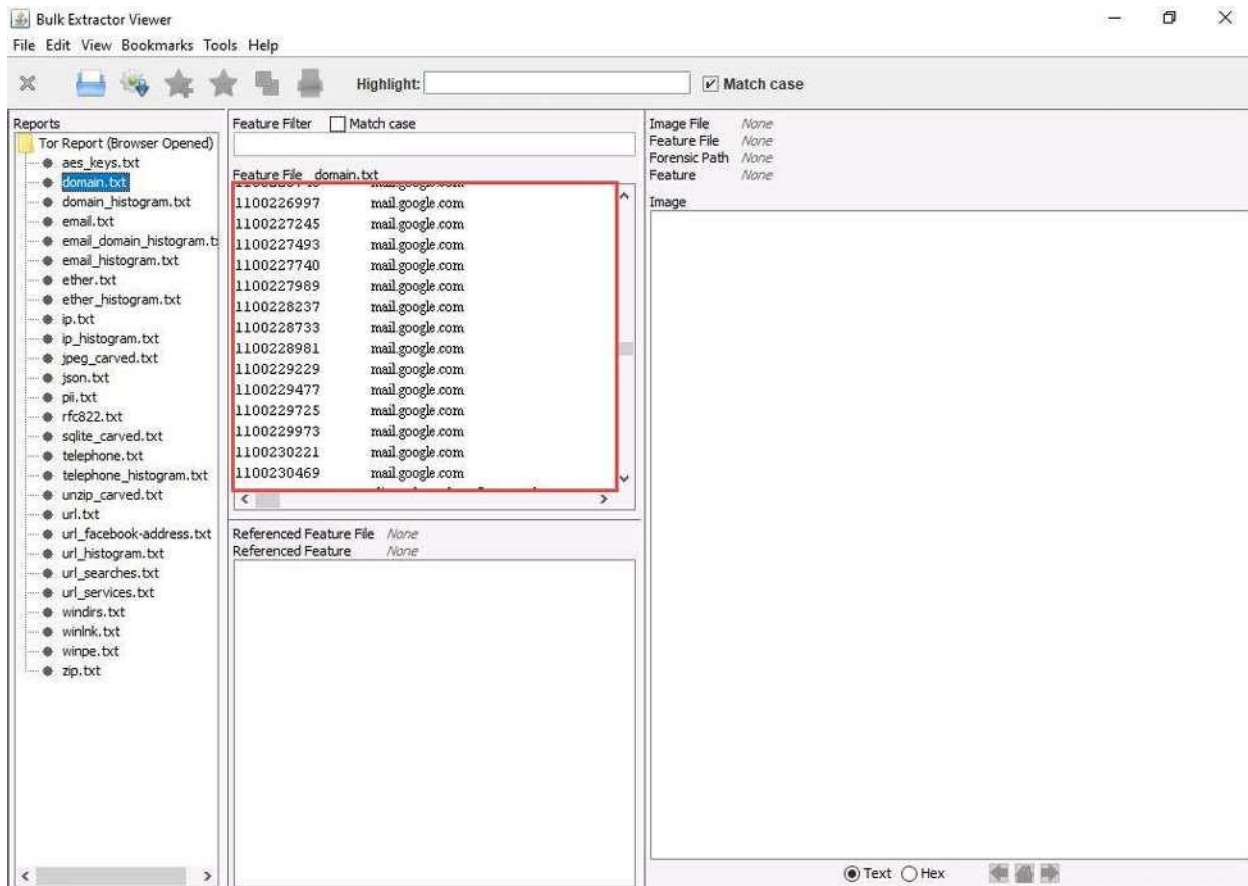


6. Upon the successful completion of the scan, close the bulk_extractor Scan window and go back to the Bulk Extractor Viewer window. We will now begin investigating the Tor Browser artifacts that were obtained when the browser was in an open state.

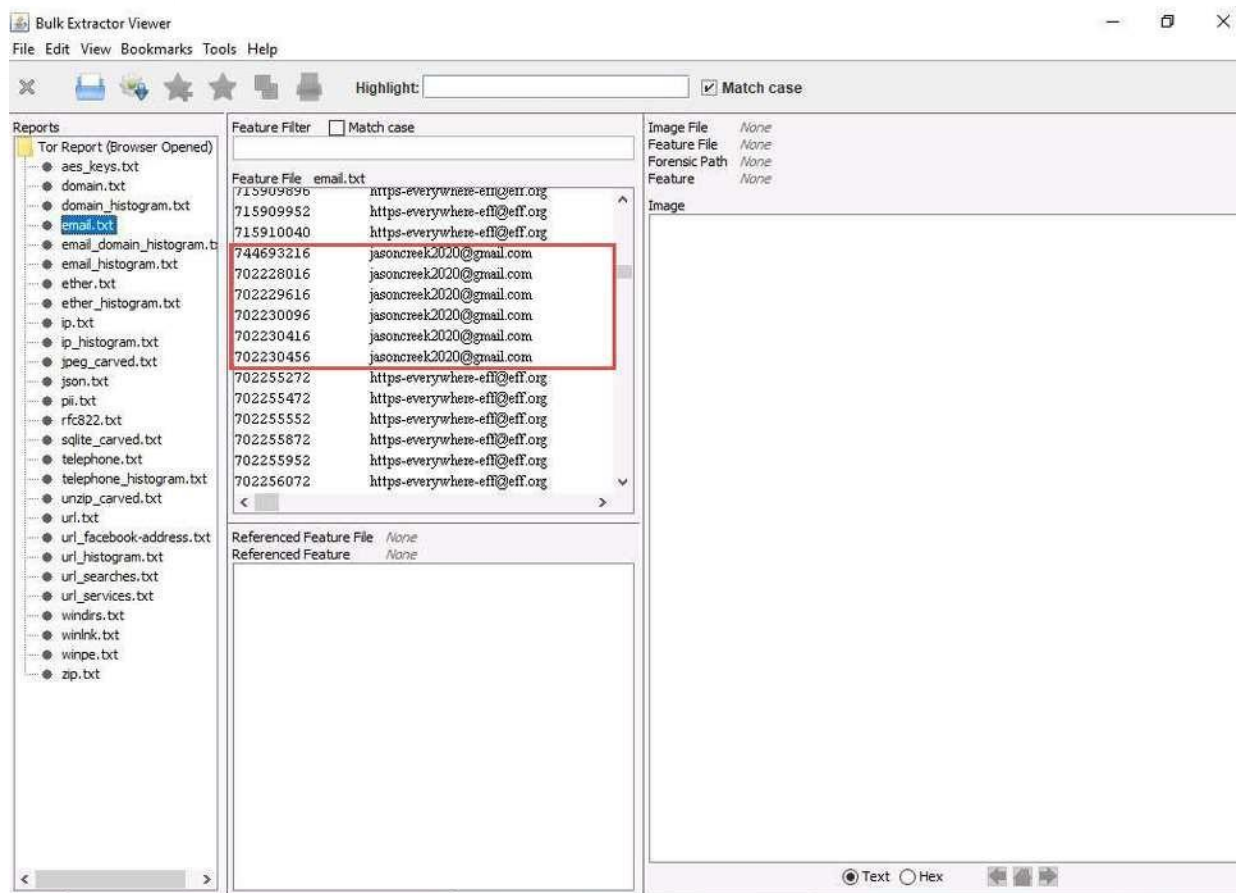
7. Now, in the left pane of the application window, you will see the Tor Report (Browser Opened) folder populated under the Reports section. Click the folder to expand it and view its contents.



8. Select the domain.txt file to determine all website domains that were visited on the suspect machine's Tor Browser. You will see several different domains listed under domain.txt. Upon scrolling down, we find numerous instances of the use of the mail.google.com domain, as seen in the screenshot below. This tells us that there were numerous instances of Gmail being used to exchange emails.



9. Now, we will look for email IDs associated with Gmail as several instances of mail.google.com under domain.txt have been located, as seen in the screenshot above. To find the email IDs that have been recorded in this memory dump file, i.e., TOR_Opened.mem, click on email.txt in the left pane under the Reports section. You will see all email IDs (including Gmail IDs) recorded on the memory dump, as highlighted in the screenshot below:



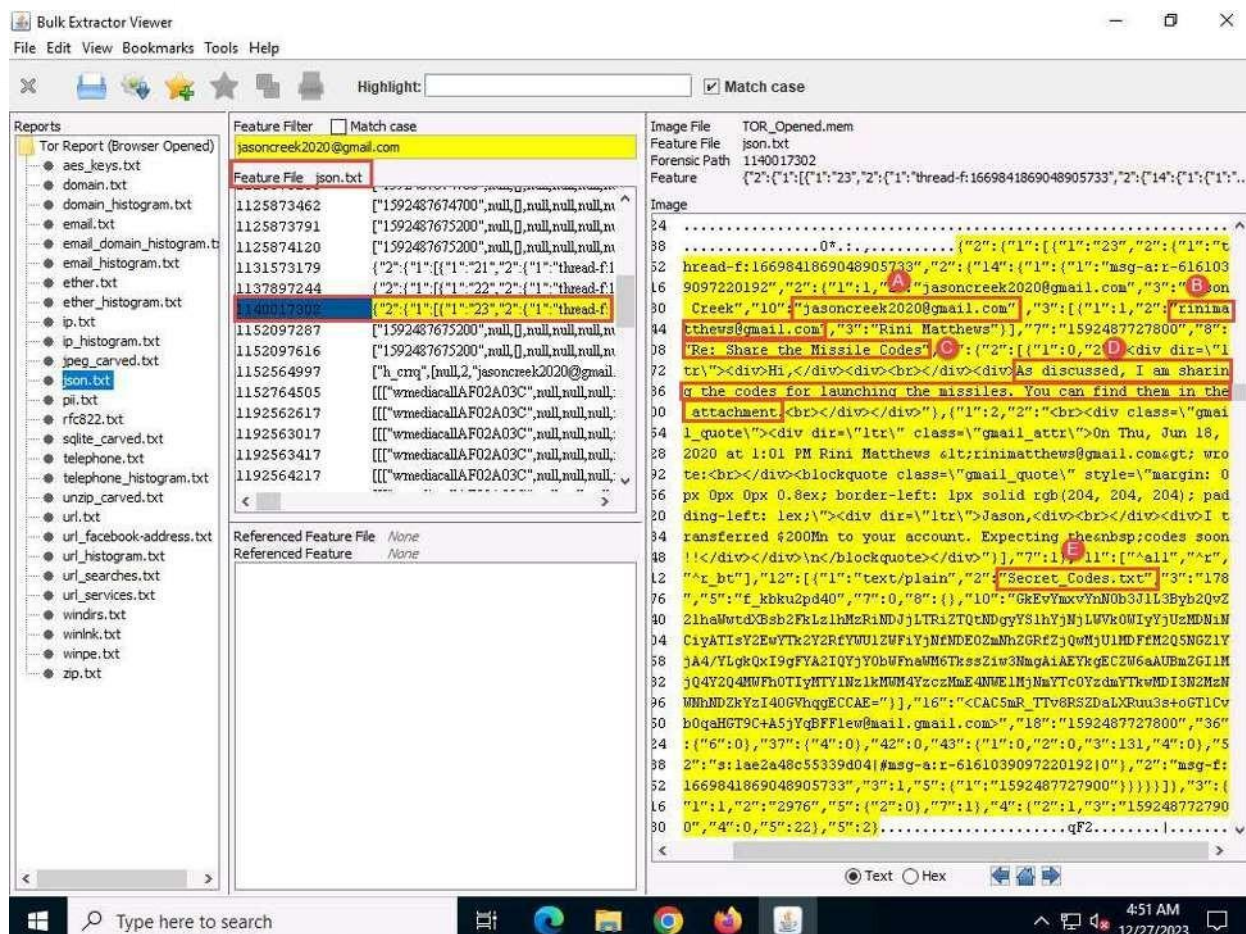
10. From the screenshot above, we can infer that there are multiple instances of the use of a Gmail ID. For demonstrative purposes, we have highlighted the region where we see the Gmail ID jasoncreek2020@gmail.com in this lab. In real-time, you might find instances wherein several email IDs from Gmail or any other email service provider(s) have been recorded.

11. Now, we will examine the contents of json.txt file. A JSON file stores information on the data exchange that has taken place between a browser/web application and a server. By examining the contents of the json.txt file here, we can retrieve the details of email exchanges on the browser (in this case, Tor Browser).

12. Therefore, when you first click on json.txt (1), you will find several entries in the Feature File section in the upper half of the middle pane in the application window. Since we have found a number of entries pertaining to the email ID jasoncreek2020@gmail.com previously under email.txt, we are assuming that email ID belongs to a suspect user. As a second step, enter jasoncreek2020@gmail.com in the Feature Filter field (2) and press Enter to obtain the artifacts of email communication related to the mentioned email ID under the Feature File section (3).

We need to carefully examine each of the artifacts of malicious email communication as seen in the screenshot above. Upon carefully examining each artifact of malicious email communication, when you click on any entry, you can see the details pane of the window.

right pane of the window.



For demonstrative ease in this lab, and to save time, we have confined our investigation to retrieving malicious email artifacts that relate to the email ID jasoncreek2020@gmail.com (assuming it to be a suspect email ID). In real-time, however, you might have to examine email communication from several other email IDs to identify the suspects or the suspect email IDs in a case.

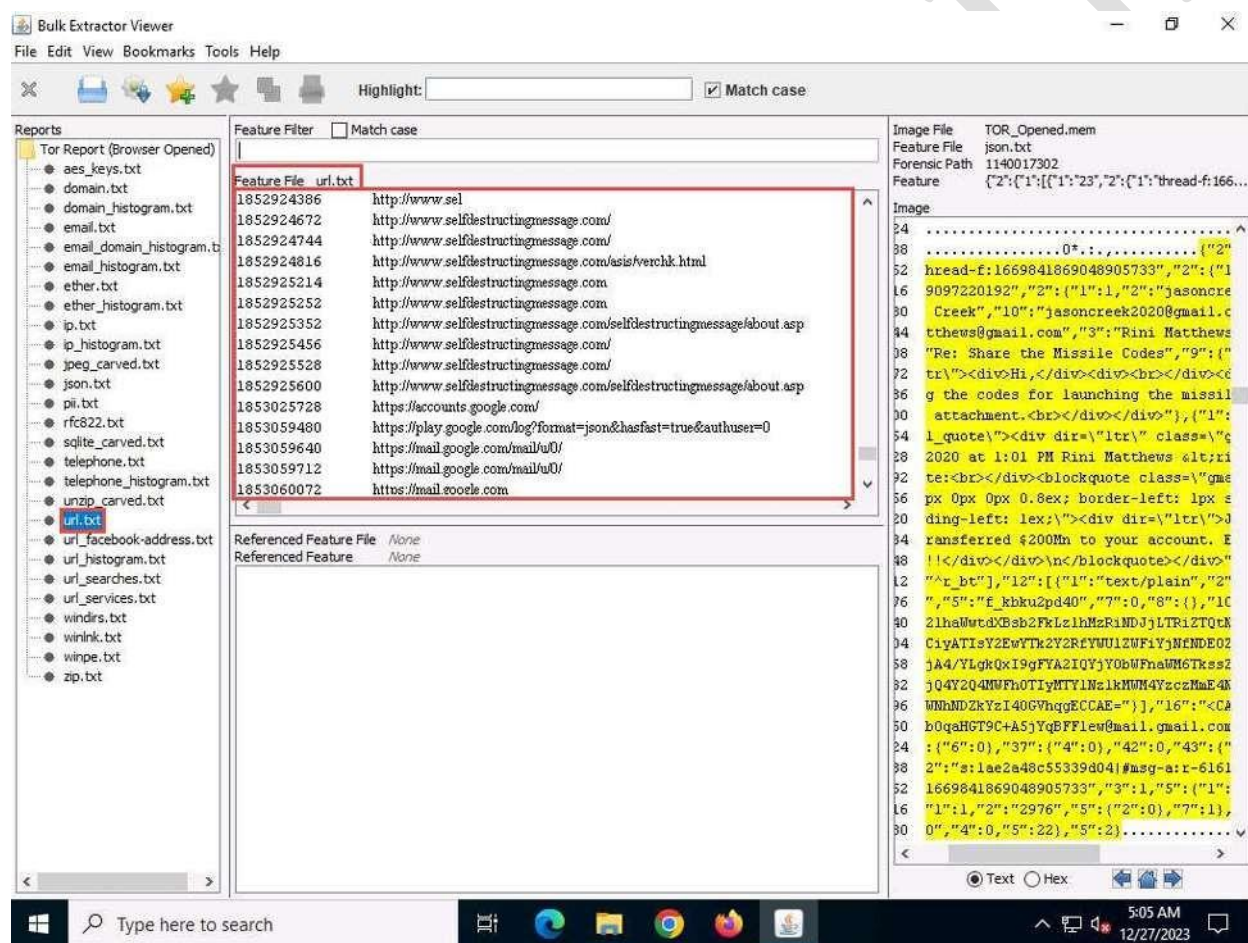
14. From the above screenshot, we can summarize our observations as follows:

- A. Email ID of the sender: jasoncreek2020@gmail.com.
- B. Email ID of the receiver: rinimatthews@gmail.com.
- C. Subject of the email: Share the Missile Codes.
- D. The body of the email: As discussed, I am sharing the codes for launching the missiles. You can find them in the attachment.
- E. Attachment found in the email: Secret_Codes.txt.

15. In this manner, we can retrieve the malicious email exchanges that took place through the Tor Browser.

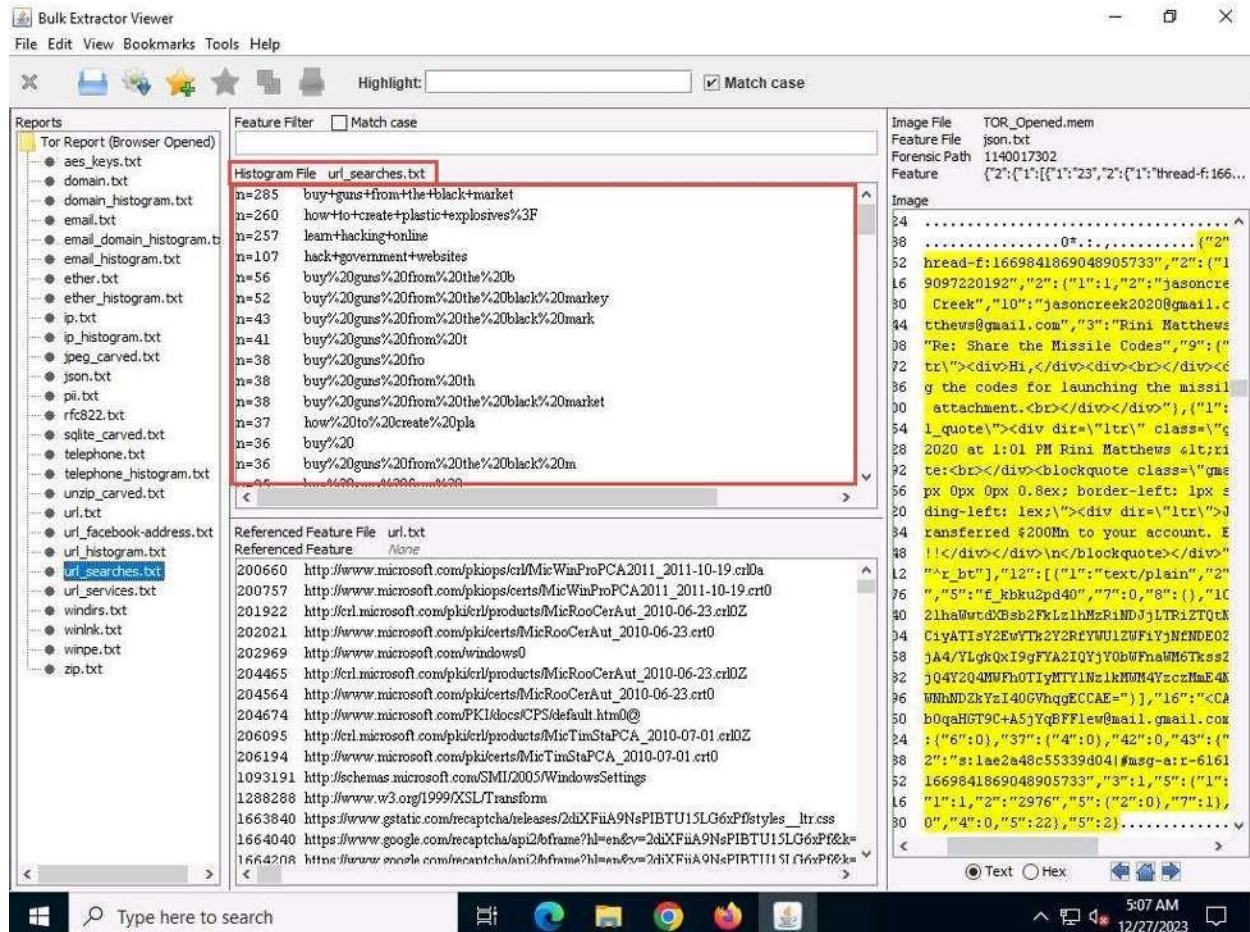
16. Now, we will retrieve the artifacts stored in url.txt file. The url.txt will provide us information on all URLs that have been visited through the suspect machine's Tor Browser. Before moving to this task, ensure to remove the jasoncreek2020@gmail.com filter from the Feature Filter field, which we had applied for our previous task of finding email artifacts from the json.txt file.

17. Now, click on url.txt in the left pane of the application window under the Reports section. The application will display all artifacts stored in url.txt under the Feature File section, as displayed in the following screenshot:



18. Now, we will examine the artifacts stored in url_searches.txt. Examining the artifacts of url_searches.txt will provide us information about all the search queries that were made on the suspect machine's Tor Browser.

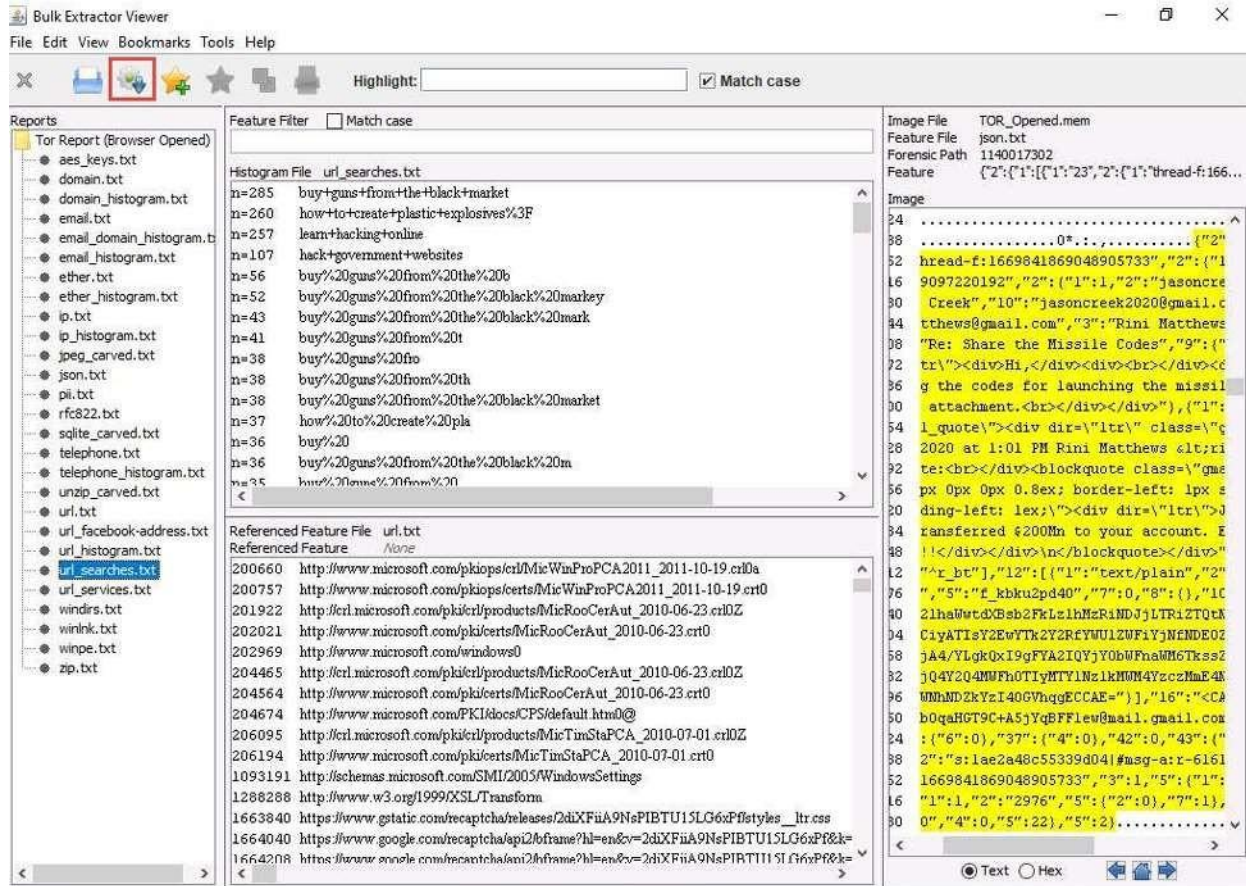
19. Click on `url_searches.txt` in the left pane of the application window under the Reports section. All queries that have been searched on the suspect machine's Tor Browser will now be listed under the Histogram File section in the upper half of the middle pane in the application window, as indicated in the screenshot below:



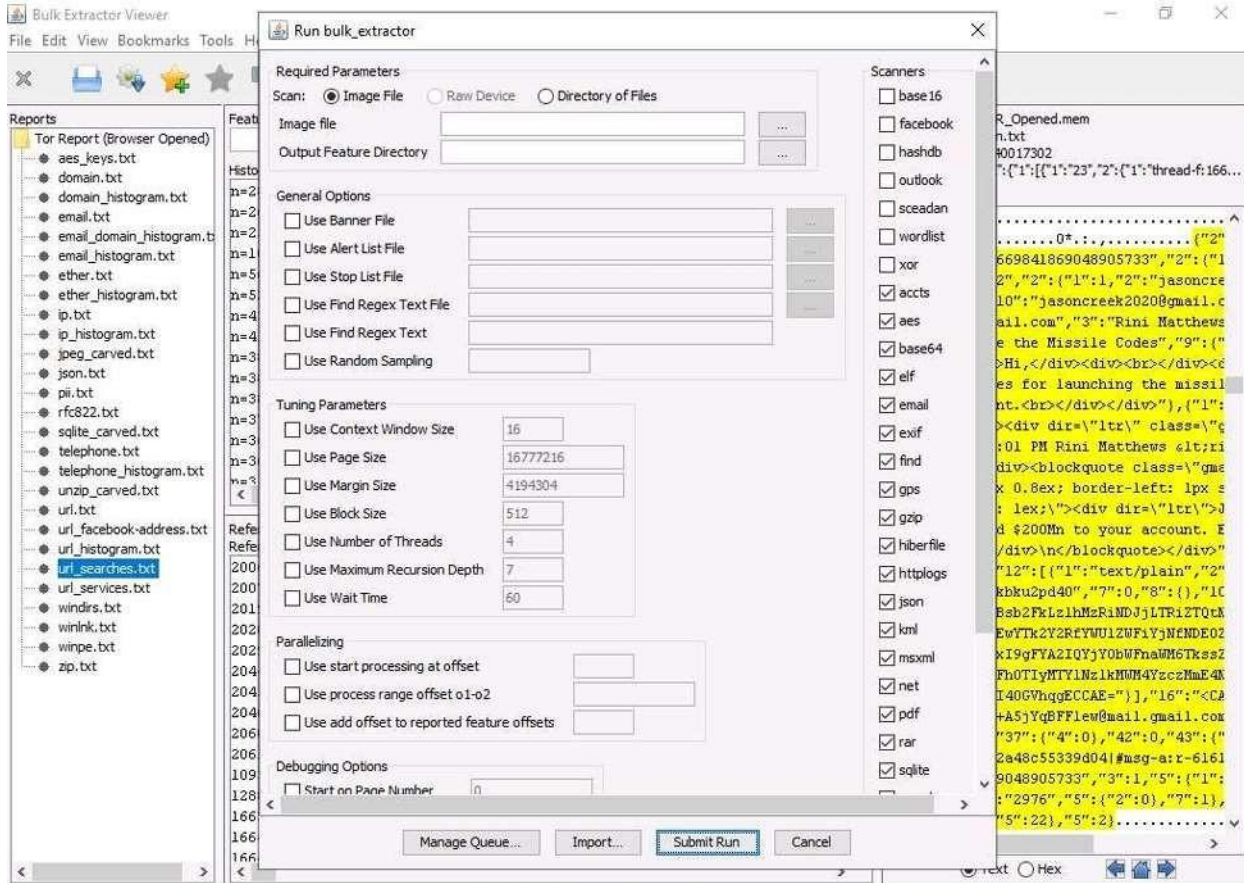
20. In this manner, we can find URLs or content that have been browsed on the suspect machine's Tor Browser.

21. Now, we will examine the Tor Browser artifacts obtained when the browser was in a closed state.

22. In the Bulk Extractor Viewer window, click on the Generate a report using `bulk_extractor` icon.

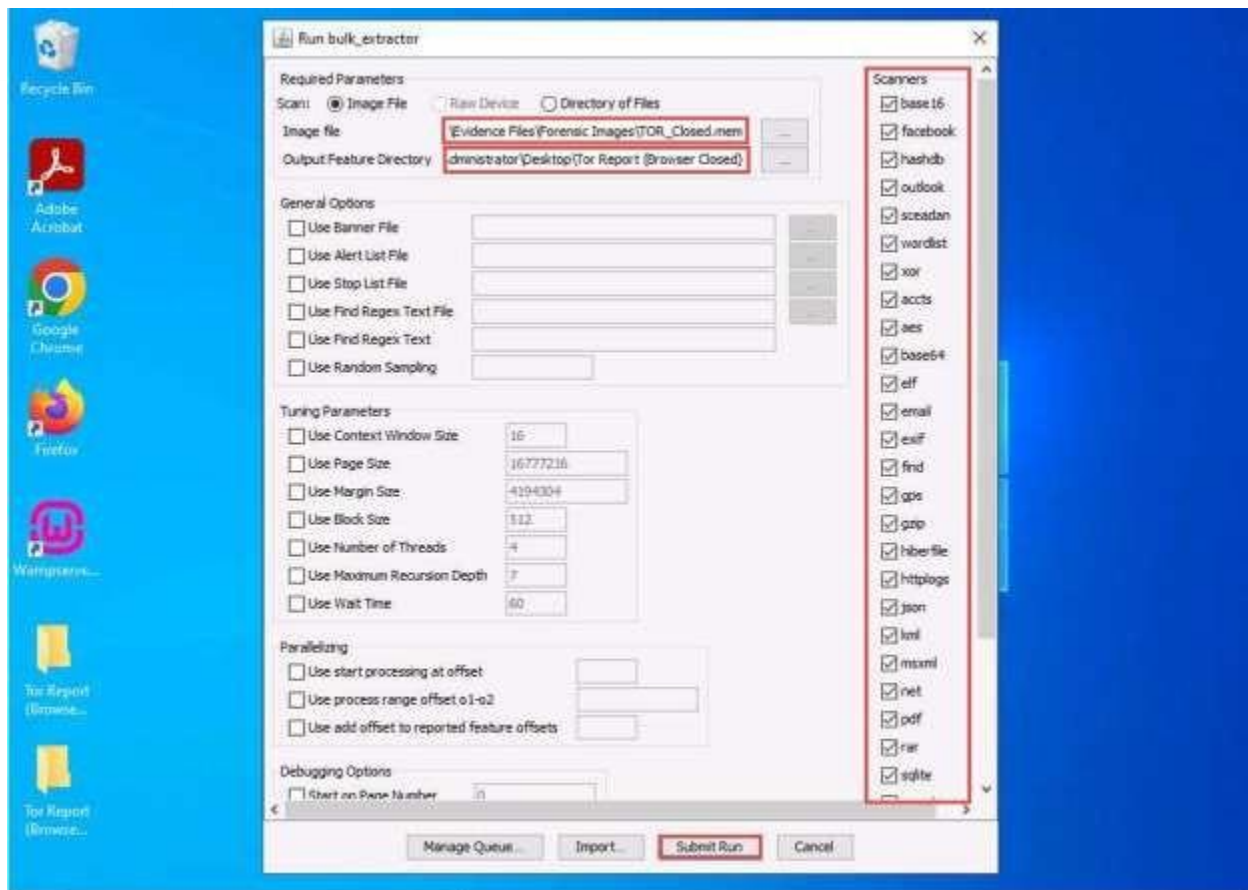


23. The Run bulk_extractor window opens, as shown in the screenshot below:



24. Now, click on the ellipsis buttons to browse and provide the Image file and the Output Feature Directory, as indicated in the screenshot below:

25. Upon clicking the ellipsis button against the Image file field, the Image File to Extract Features From window will appear. Navigate to E:\CHFI-Tools\Evidence Files\ Forensic Images. From the Files of type drop-down, select All Files, select the file TOR_Closed.mem, and then click Open to provide the Image File.



26. We will begin our forensic examination by first retrieving the information stored in domain.txt.

27. Select the domain.txt file to determine all website domains that were visited on the suspect machine's Tor Browser. You will see several different domains listed under the Feature File domain.txt section in the middle pane of the window. Upon scrolling down, we find numerous instances of the use of the mail.google.com domain, as seen in the screenshot below. This tells us about numerous instances where Gmail was used to exchange emails. Click/Select any of the instances of mail.google.com if you want to find any specific artifacts associated with it. The artifacts, if any, will appear in the right pane of the window.

29. Close all open windows.